

TADHG MCDONALD-JENSEN

Seeking full product life-cycle development in an agile work environment.

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Full Stack Developer

Ericsson 📍 Kanata, ON

Backend Developer

May 2018 – Dec 2018

- Worked with radio technicians and hardware developers to measure indoor cellular deployment metrics.
- Reduced execution time of exhaustive test suite by 40% through optimizing MATLAB algorithms.
- Audited database migration plan, identified missing and redundant fields and composed a summary.
- Retro-fitted javascript codebase with strict typesafety between UI and database components.

Frontend Developer and Coordinator

Jan 2019 – Aug 2019

- Designed user interface control scheme optimizing accessibility.
- Performed task breakdown in JIRA to maximize parallel development and track dependencies.
- Coordinated 6 programmers in implementing Typescript frontend with React and Redux frameworks.
- Performed comprehensive code reviews with Gerrit and Git.
- Ensured seamless merging and integration of features.
- Wrote developer on-boarding training material.

Teaching Assistant

Carleton University 📍 Ottawa, ON

Real-Time Concurrent Systems (SYSC 3303)

Winter 2022

- Evaluated student projects written in Java, Composed detailed feedback about potential bugs or race conditions.
- Answered technical questions and clarified course logistics by email and class wide announcements.
- Recorded and published a video analysis of a thread synchronized data structure demonstrating a race condition.
- Identified and corrected a bug in professor's JUnit test code used to evaluate student submissions.

Discrete Simulation/Modeling (SYSC 4005)

Winter 2023

- Evaluated student work on functional correctness as well as whether it followed proper discrete time simulation principles.
- Used Software Similarity tooling to identify cases of plagiarism. Presented report of academic misconduct to Engineering Dean.
- Explained stochastic process statistics to students.

Operating Systems (SYSC 4001)

Fall 2023

- Explained asynchronous interrupts, process forking, and other core concepts to operating systems.
- Engaged students by using a analogy of a father-son dual murder plot to explain process interrupts.
- Helped students troubleshoot Unix VM issues across wide array of virtual machine configurations.

About Me

Team player who excels in problem solving and writing clean maintainable code

Education

M.Eng. in Computer Systems

Carleton University

📅 Sept 2022 – Dec 2025

B.Eng. in Computer Systems

Carleton University

📅 Sept 2016 – Apr 2021

10.68/12 CGPA (A-/A average)

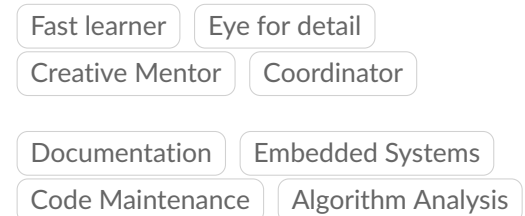
Languages

English

Français



Strengths



Achievements

📄 tadhg.mcdj.solutions/coloursort

Responsive web game written with no javascript dependencies

📈 [21k+ Reputation on StackOverflow](#)

Posted over 600 answers to help others with programming questions

🎓 **Engineering 4th year project** Wrote frontend program to connect to bio-sense wearable and read data in custom serial encoding and graph live sensor readings.

Professional Practice (ECOR 4995)

Winter 2024

- Elaborated on application of proper engineering principles.
- Evaluated student essays on subjective ethical topics.

Projects

Self Hosted Web Server

Ongoing

- Debugged and modified the u-boot sequence over serial connection to a Turris Omnia wifi router.
- Wrote custom linux system configuration to run file hosting and networking services on the router.
- Cross compiled the kernel and services code from source on my personal computer and pushed updates remotely.

Piano Halloween Costume

Oct 2022

- Write interrupt driven code in C to identify GPIO connections.
- Sent audio playback through bluetooth when circuit was closed.
- Embedded microcontroller into piano costume to play notes when keys were touched with metal leads.

Matlab Radio Cross Correlation Analyzer

Aug 2021

- Designed and implemented a GUI tool to view complex valued radio timing correlation data.
- Implemented several algorithms to estimate parameters to approximate sample data as linear combination of pulses.
- Demonstrated that complex phase analysis is necessary to determine appropriate approximation technique.

Ask me about my system clock!

Contributions to open source

- Submitted several bug fixes to eslint-typescript with unit tests ensuring 100% code coverage of new logic.
- Wrote a formal feature proposal for Typescript to address one of the longest standing open issues with the language.
- Identified solution to an obscure bug in the core python interpreter.
- Submitted patch to Logisim enabling drag-n-drop functionality.

Reference

Alexander Craig

Product Development Leader, Ericsson

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Both students and on-boarding mentees have commended me for my clarity and approachability.